

HubSpot's personalization feature allows you to tailor your emails at scale by inserting dynamic variables—called personalization tokens—that automatically pull in data from your CRM for each recipient. Here's how it works in practice, with a focus on using someone's first name in mass emails:

How Personalization Tokens Work

- **Dynamic Placeholders:** You insert personalization tokens (like `{{ contact.firstName }}` or simply `FIRSTNAME` in the email builder) wherever you want recipient-specific information to appear³⁶.
- **CRM Integration:** When you send an email, HubSpot checks each recipient's profile in your CRM and substitutes the token with the actual value stored for that contact (e.g., "John," "Sarah," etc.)³⁶.
- **Default Values:** If a contact does not have a value for a property (e.g., first name is missing), you can set a default value (like "Customer" or "Friend") as a fallback. This ensures that the email still looks personalized, even if some data is missing²⁷.

Example: Using First Name in Mass Emails

Imagine you want to send a mass email starting with "Hi [First Name],":

1. **Insert the Token:** In your email editor, place your cursor where you want the first name to appear. Select the personalization token for first name (usually `{{ contact.firstName }}` or `FIRSTNAME`)³⁷.
2. **Set Default Value:** If you want, set a default value (e.g., "Customer") to be used if a recipient does not have a first name in your CRM².
3. **Send to a List:** Select your target list (active or static) and send the email. HubSpot automatically replaces the token with each recipient's first name, or the default if the field is empty³.
4. **Result:** Each recipient receives an email that starts with their own name, making the message feel individualized even though it was sent to many people³⁶.

Additional Features

- **Smart Content:** You can go beyond just names and use smart content rules to show or hide different email sections based on recipient properties (like lifecycle stage, location, or list membership)³⁶.
- **Custom Properties:** You can create and use tokens for any custom contact or company property in your CRM⁴⁶.

- From Name Personalization: You can even personalize the sender's name or email address by using tokens for properties like contact owner, making emails feel more personal and relevant⁵.

Summary Table

Feature	Description
Personalization Tokens	Insert dynamic placeholders (e.g., <code>{{ contact.firstName }}</code>) in emails
CRM Integration	Tokens are replaced with actual values from each recipient's CRM profile
Default Values	Set fallback text if a property is missing
Smart Content	Show/hide email sections based on recipient data
Custom Properties	Use tokens for any custom field in your CRM

Key Point

HubSpot's personalization feature ensures that every recipient receives a message that feels individually crafted, even when sending mass emails, by dynamically inserting their own information from your CRM

To create a custom CRM personalization feature similar to HubSpot's, developers need to implement dynamic token replacement, CRM data integration, and scalable email delivery. Below is a structured guide for developers:

Core Components to Implement

1. Personalization Token System
 2. CRM Data Integration
 3. Email Templating with Dynamic Content
 4. Fallback Handling for Missing Data
-

1. Personalization Token Syntax

Define token syntax (e.g., `{{ contact.firstName }}`) and create a parser to identify tokens in email templates.

Example Code (Python):

python

```
import re

def replace_tokens(content, contact_data):
    tokens = re.findall(r'\{\{\s*(\w+\.|\w+)\s*\}\}', content)
    for token in tokens:
        property_type, property_name = token.split('.')
        value = contact_data.get(property_name, "Customer") #
        # Fallback to "Customer"
        content = content.replace(f'{{{token}}}', value)
    return content
```

2. CRM Data Integration

Connect to your CRM database to fetch contact properties.

Steps:

- Create a `contacts` table with fields like `first_name`, `email`, and `subscription_status`.
- Use SQL or an ORM to query data:

python

```
def fetch_contacts():
    # Example using SQLAlchemy
    contacts = db.session.query(Contact).filter(Contact.opt_in ==
True).all()
    return [{"firstName": c.first_name, "email": c.email} for c in
contacts]
```

3. Email Templating Workflow

Process Flow:

1. Insert tokens into email templates (e.g., "Hi {{ contact.firstName }}").
2. For each recipient:
 - Fetch their data from CRM.
 - Replace tokens with actual values.
 - Use a default (e.g., "Customer") if data is missing.

Example Email Sending (SendGrid):

python

```
from sendgrid import SendGridAPIClient
from sendgrid.helpers.mail import Mail

def send_email(to_email, personalized_content):
    message = Mail(
        from_email="noreply@yourcrm.com",
        to_emails=to_email,
        subject="Your Personalized Email",
        html_content=personalized_content
    )
```

```
try:
    sg = SendGridAPIClient(os.getenv("SENDGRID_API_KEY"))
    response = sg.send(message)
except Exception as e:
    print(f"Error: {e}")
```

4. Advanced: Dynamic Content via Feature Flags

Implement rules to show/hide content based on user attributes (e.g., premium vs. regular users).

Example (LaunchDarkly Integration):

python

```
import ldclient

def evaluate_feature_flag(user_id, user_attributes):
    context = Context.builder(user_id) \
        .set("email", user_attributes["email"]) \
        .set("subscription_status",
user_attributes["status"]) \
        .build()
    return ldclient.get().variation("show-premium-content",
context, False)
```

Developer Documentation Outline

text

CRM Personalization Feature Guide

1. Token System

*- **Syntax**: Use `{{ contact.propertyName }}` in templates.*

- **Parser**: Regex-based token replacement [see code snippet].

2. Database Integration

- **Schema**: Ensure `contacts` table includes fields for personalization.
- **Query**: Fetch contacts with opt-in status.

3. Email Engine

- **Template Setup**: Use tokens in HTML/plain text templates.
- **Fallbacks**: Configure defaults in `replace\_tokens()`.

4. Feature Flags (Optional)

- **Integration**: Use LaunchDarkly SDK for dynamic content rules.

Testing Checklist

- Validate token replacement for:
 - Complete CRM data
 - Missing data (fallback triggers)
 - Special characters in properties

Security Considerations

- **Data Sanitization**: Escape HTML in user-provided fields to prevent injection.
- **API Keys**: Store email service credentials in environment variables.
- **GDPR Compliance**: Ensure opt-in status is respected before sending emails.

For additional reference, see HubSpot's [personalization token documentation](#) and LaunchDarkly's [feature flag examples](#)

[Here is HubSpot Token documentation](#)

Personalize your content

Last updated: May 16, 2025

Available with any of the following [subscriptions](#), except where noted:

 All products and plans

.With personalization tokens, you can show personalized content to your contacts based on their property values. For example, you could address an email to "Dear [first name]," and pull in each contact's first name from their contact record.

This article covers how to add personalized content to rich text modules in the content editor, marketing email editor, SMS editor, and design manager. Learn how to add personalization to [email subject lines](#) and [sales email templates or snippets](#).

Before you get started

Before you begin working with this feature, make sure to fully understand what steps should be taken ahead of time, as well as the limitations of the feature and potential consequences of using it.

Understand requirements

- A **Marketing Hub** subscription is required to add personalization tokens to forms, SMS messages, and marketing automation.
- A **Content Hub** subscription is required to add personalization tokens to website and landing pages.
- The [Marketing SMS add-on](#) or [Marketing SMS Short Code add-on](#) is required to add personalization tokens to SMS messages.

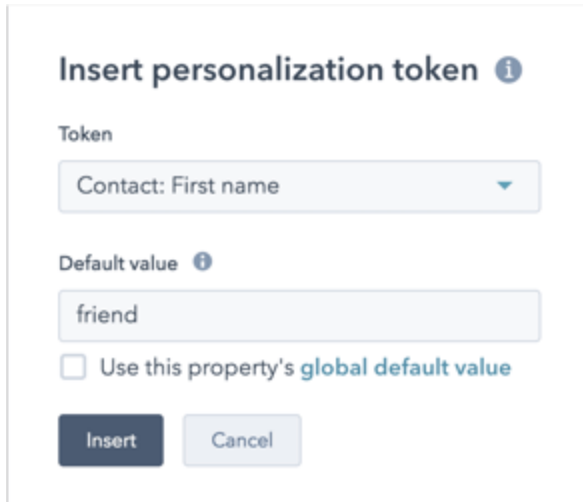
Understand limitations & considerations

- In order for personalization tokens to show a value, the contact viewing the token must be a [tracked visitor](#) with a contact record in your CRM and a known property value for that token.

- If you don't set a default value for a personalization token, the token will be left blank for contacts who don't have a value for that property.
- Only contact and company personalization tokens can be tested in [test emails](#).
- [Custom object](#) names that contain hyphens cannot generate personalization tokens. Learn more about [editing custom object names](#).
- In quotes, personalization tokens should not be used for properties updated after a quote is published, such as Payment status and Payment date. These properties will not update after the quote is published.

Add personalization to your content

- Navigate to your content:
 - **Website Pages:** In your HubSpot account, navigate to **Content > Website Pages**.
 - **Landing Pages:** In your HubSpot account, navigate to **Content > Landing Pages**.
 - **Email:** In your HubSpot account, navigate to **Marketing > Email**.
 - **SMS:** In your HubSpot account, navigate to **Marketing > SMS**.
- Click the **name** of your content.
- In the editor, click a **rich text module**, then click **Personalize**.
- Click the **Type** dropdown menu and select a **category** of personalization token.
- Click the **Token** dropdown menu and select a **property**. You may be prompted to select additional information about your chosen property.
- Enter a **fallback value** for the personalization token. The fallback value will be used when the contact does not have a value for that specific property. Learn more about [setting global default values for personalization tokens](#).
- Click **Insert**. You'll see the token highlighted in the rich text module with the name of the selected property.



Insert personalization token ⓘ

Token

Contact: First name ▼

Default value ⓘ

friend

☐ Use this property's [global default value](#)

Insert Cancel

Review available personalization tokens

The available tokens and their associated properties depend on the type of content you're creating.

Personalization tokens available for all content types

All content types can include personalization tokens that use a company or contact property.

- **Company:** these tokens use the values for the [company properties](#) associated with contacts in your HubSpot database. This includes tokens for company owner, total revenue, industry, and name.
- **Contact:** these tokens use the values for any [contact properties](#) associated with contacts in your HubSpot database. This includes tokens for contact owner, first name, last name, and email.

Personalization tokens available for marketing emails

The personalization tokens available in a marketing email depend on the type of email you're creating:

- **All marketing emails:** if you're editing a [regular marketing email](#) or [automated marketing email](#), you can include personalization tokens related to your office

location details or email subscription type:

- **Office Location:** these tokens are from the [email footer information](#) in your settings. This includes tokens for city, state, and zip code.
- **Subscription type:** this token uses the name of the email's [subscription type](#).
- **Automated marketing emails:** if you have a **Marketing Hub** Professional or Enterprise account, and you have [Edit access to workflows](#), you can insert a personalization token into an automated marketing email that matches the type of record that will be enrolled in your workflow:
 - **Deal:** these tokens use the values for the [deal properties](#). This includes tokens for the deal owner, amount, and closed won date.
 - **Ticket:** if you have a **Service Hub** Starter, Professional, or Enterprise account, you can use [ticket properties](#).
 - **Custom object:** if you have an Enterprise account, you can use properties from any [custom objects](#) you've created.
 - **Subscription:** use a property from a [subscription](#) record associated with the recipient.
 - **Invoice:** use a property from an enrolled [invoice](#) record.
 - **Quote:** use a property from an enrolled [quote](#) record.
 - **Cart:** use a property from an enrolled cart record. Carts can be synced from one of your ecommerce accounts using data sync:
 - In your HubSpot account, navigate to **CRM > Contacts**.
 - Click the **Contacts** dropdown menu, then select **Carts**.
 - Click **Set up data sync**, then follow the prompts to [connect HubSpot and Shopify using data sync](#).
 - **User:** use a property from an associated user in your HubSpot account.

Preview personalization

Once you've added personalization tokens to your content, you can preview how that content would appear to a specific contact.

Preview personalization in pages

- *Navigate to your content:*
 - **Website Pages:** *In your HubSpot account, navigate to **Content** > **Website Pages**.*
 - **Landing Pages:** *In your HubSpot account, navigate to **Content** > **Landing Pages**.*
- *Click the **name** of your page.*
- *In the content editor, click **Preview** in the top right.*
- *In the top left, click the **Preview as** dropdown menu and select **Contact**.*
- *Click the second **dropdown menu** and select a **contact**.*

Preview personalization in emails

- *In your HubSpot account, navigate to **Marketing** > **Email**.*
- *Click the **name** of a draft or automated email.*
- *In the email editor, click the **Preview** dropdown menu in the top right, then select **Preview as contact**.*
- *On the preview screen, click the **Preview as a specific contact** dropdown menu and select a **contact**.*

Preview personalization for SMS

- *In your HubSpot account, navigate to **Marketing** > **SMS**.*
- *Click the **name** of a draft SMS message.*
- *In the SMS editor, click the **View default preview message** on the right and select a **contact**. The preview below will update to show how the message would appear for that contact.*

Add personalization to a coded template

You can use [supported variables](#), [HubL filters](#), and [HubL functions](#) to add personalization tokens to a coded template.

- *In your HubSpot account, navigate to **Content** > **Design Manager**.*
- *In the left column, click the **name** of a [coded template](#).*
- *In the code of the template, add the HubL variable surrounded by two sets of brackets where you want the personalization token to appear.*

- *For example, if you wanted to add the First name contact property to a coded template, you would format it like this: Zohe.*
- *When you've finished editing your template, click **Publish changes** in the top right.*